

AIRPORT RUNWAY MANAGEMENT SIMULATOR

Week 5 Project Report

Runway Allocation System

1. Introduction

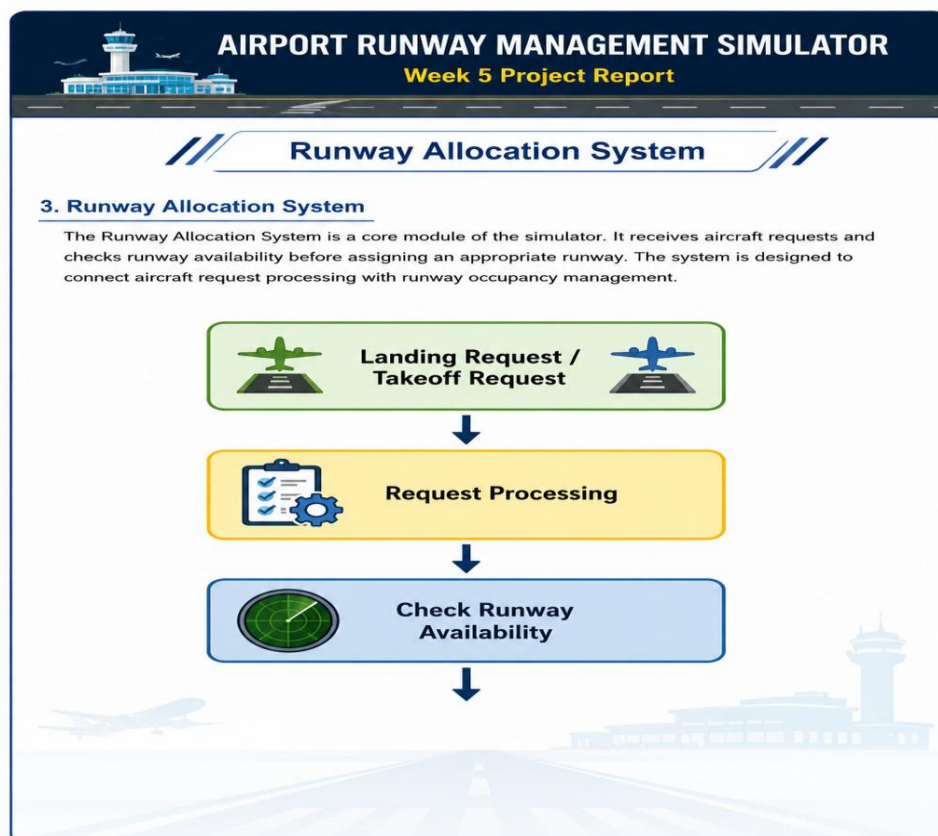
The Airport Runway Management Simulator is a Java-based software project designed to simulate basic airport runway management activities. In Week 5, the project focuses on developing the Runway Allocation System. The main objective is to automatically assign available runways, process landing and takeoff requests, and manage runway occupancy.

2. Week 5 Objectives

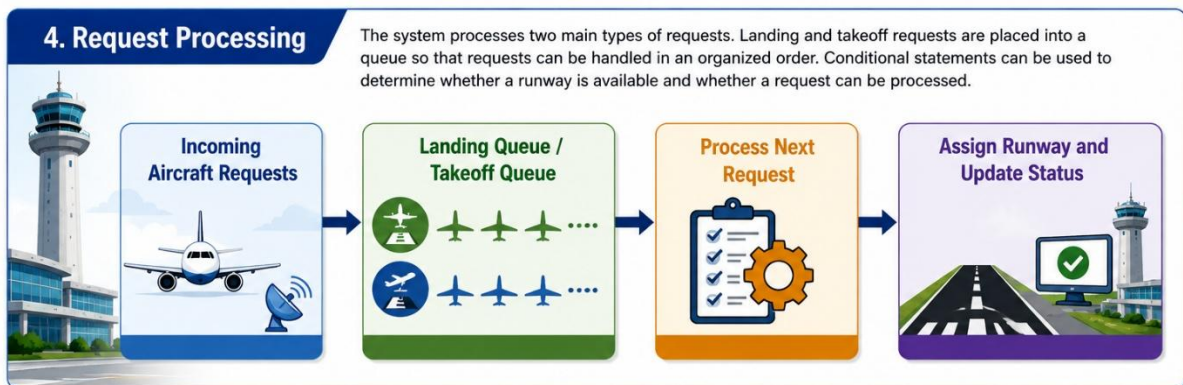
The main objectives of Week 5 are:

- Assign runways automatically.
- Process landing requests.
- Process takeoff requests.
- Manage runway occupancy.

3. Runway Allocation System



4. Request Processing



5. Runway Occupancy Management



6. Technologies and Concepts to Learn

The Week 5 implementation focuses on the following concepts:

- Queue Data Structure — to organize landing and takeoff requests.
- Conditional Statements — to check runway availability and request conditions.
- Event Driven Programming — to respond to request and runway status events.

7. GitHub Upload

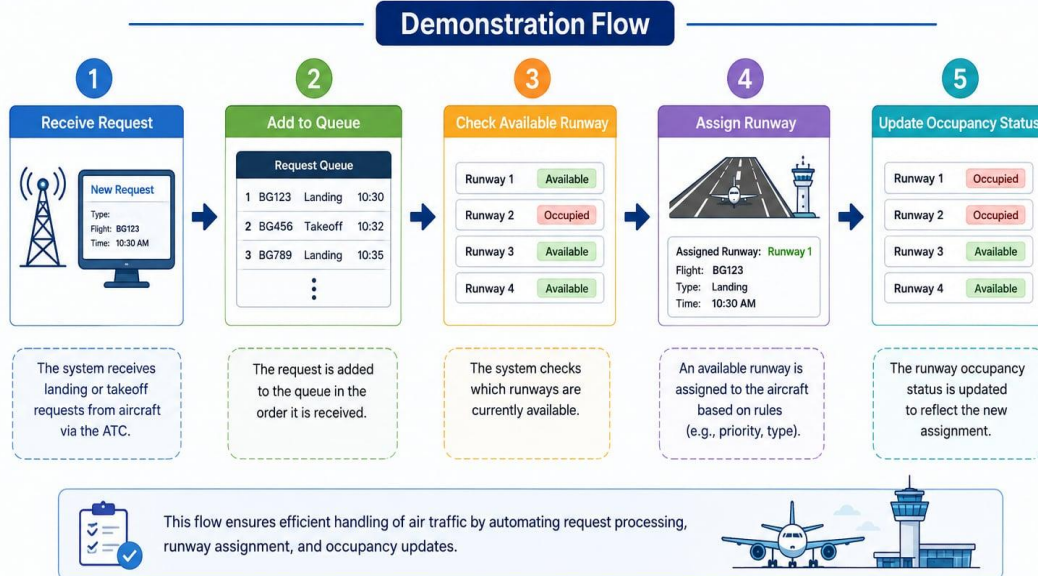
The Week 5 implementation will be uploaded to the project GitHub repository. The required materials are:

- Runway Allocation Module
- Request Processing Module
- Related Java source files required to run the modules.

8. Demonstration

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At the end of Week 5, the project should demonstrate automatic runway assignment and request processing. The system should show how landing and takeoff requests are received, queued, processed, assigned to available runways, and how runway occupancy is updated.



9. Expected Outcome

After completing Week 5, the Airport Runway Management Simulator will have a basic Runway Allocation System. The system will be able to process landing and takeoff requests, assign available runways automatically, and manage runway occupancy. This module will provide an important connection between aircraft requests and airport runway operations.

10. Conclusion

Week 5 focuses on developing the Runway Allocation System of the Airport Runway Management Simulator. By using queues, conditional statements, and event-driven programming concepts, the project will be able to process aircraft requests in an organized manner. The completed module will support automatic runway assignment and occupancy management and prepare the project for further development.